

☒ Easy (10 Questions)

1.

What is the primary goal of ethical hacking?

- A) To hack systems without permission
- B) To find vulnerabilities to improve security
- C) To create malware
- D) To crash servers

Answer: B

Explanation: Ethical hacking is done to identify security weaknesses with permission.

2.

Which type of ethical hacking involves no prior knowledge of the target system?

- A) White Box
- B) Black Box
- C) Gray Box
- D) Blue Team

Answer: B

Explanation: Black Box testing simulates an external attacker with no internal info.

3.

What does footprinting primarily involve?

- A) Crashing a server
- B) Gathering information about a target system
- C) Writing exploits
- D) Installing malware

Answer: B

Explanation: Footprinting is the reconnaissance phase to collect data on the target.

4.

Which tool is commonly used to trace the path packets take to a host?

- A) Ping
- B) Traceroute
- C) Nmap
- D) Wireshark

Answer: B

Explanation: Traceroute shows intermediate hops in network path.

5.

Port scanning is used to find:

- A) Open ports on a host
- B) Network devices
- C) Vulnerabilities in software
- D) Firewall configurations

Answer: A

Explanation: Port scanning checks which ports are open for communication.

6.

Which scan sends a SYN packet but does not complete the TCP handshake?

- A) NULL Scan
- B) FIN Scan
- C) SYN Scan
- D) XMAS Scan

Answer: C

Explanation: SYN scan is a half-open scan; it sends SYN but doesn't complete connection.

7.

Social engineering attacks exploit:

- A) Software vulnerabilities
- B) Network protocols
- C) Human psychology
- D) Hardware flaws

Answer: C

Explanation: Social engineering tricks people into revealing confidential info.

8.

Vulnerability scanning differs from port scanning because it:

- A) Checks for open ports
- B) Detects software weaknesses and misconfigurations
- C) Maps network devices
- D) Sends ping requests

Answer: B

Explanation: Vulnerability scanners look for security issues beyond just open ports.

9.

Which scanning method tries to avoid detection by firewalls?

- A) Active scanning
- B) Passive scanning
- C) Stealth scanning
- D) Network scanning

Answer: C

Explanation: Stealth scanning attempts to avoid triggering alerts.

10.

Which of the following is NOT a type of ethical hacking?

- A) Black Box Testing
- B) White Box Testing
- C) Gray Box Testing
- D) Blue Box Testing

Answer: D

Explanation: Blue Box testing is not a recognized ethical hacking type.

☒ Medium (20 Questions)

11.

What is a security evaluation plan used for?

- A) To launch attacks
- B) To document scope and rules of an ethical hacking engagement
- C) To create malware
- D) To fix vulnerabilities

Answer: B

Explanation: It defines the scope, objectives, and guidelines for security testing.

12.

Which of the following is a passive reconnaissance technique?

- A) Ping sweep
- B) Dumpster diving
- C) Port scanning
- D) SYN scan

Answer: B

Explanation: Dumpster diving involves gathering info without network interaction.

13.

Which flag combination is used in an XMAS scan?

- A) SYN only
- B) FIN, URG, PSH
- C) No flags
- D) ACK only

Answer: B

Explanation: XMAS scan sends FIN, URG, and PSH flags.

14.

What does a NULL scan do?

- A) Sends packets with all flags set
- B) Sends packets with no TCP flags set
- C) Sends only SYN packets
- D) Sends only ACK packets

Answer: B

Explanation: NULL scan sends TCP packets with no flags set.

15.

In an IDLE scan, what is the purpose of the “zombie” host?

- A) To scan ports on the target
- B) To hide the attacker’s IP address
- C) To crash the target
- D) To defend the network

Answer: B

Explanation: The zombie sends spoofed packets, masking the attacker.

- 16.
- Which scanning technique is less likely to be detected by Intrusion Detection Systems (IDS)?**
- A) SYN scan
 - B) Full connect scan
 - C) Stealth scan
 - D) Ping scan
- Answer: C**
- Explanation:** Stealth scans attempt to evade IDS detection.
- 17.
- What is the difference between network scanning and port scanning?**
- A) Network scanning finds active hosts; port scanning finds open ports on hosts
 - B) Both are the same
 - C) Port scanning finds network devices
 - D) Network scanning finds open ports
- Answer: A**
- Explanation:** Network scanning detects hosts; port scanning checks ports on those hosts.
- 18.
- Which type of ethical hack simulates an insider threat?**
- A) Black Box
 - B) White Box
 - C) Gray Box
 - D) None of the above
- Answer: B**
- Explanation:** White Box testing assumes full internal knowledge.
- 19.
- Which tool is commonly used for port scanning?**
- A) Wireshark
 - B) Nmap
 - C) Metasploit
 - D) John the Ripper
- Answer: B**
- Explanation:** Nmap is a popular port scanner.
- 20.
- Footprinting is generally performed during which phase of ethical hacking?**
- A) Exploitation
 - B) Reconnaissance
 - C) Reporting
 - D) Post-exploitation
- Answer: B**
- Explanation:** Footprinting is part of reconnaissance.
- 21.
- Social engineering attacks can be prevented primarily by:**
- A) Patching software

- B) User awareness and training
- C) Firewall installation
- D) Encryption

Answer: B

Explanation: Educating users is key to defending against social engineering.

22.

Which of the following is true about SYN scans?

- A) They complete the TCP handshake
- B) They are noisy and easily detected
- C) They are half-open scans that don't complete the handshake
- D) They send no packets

Answer: C

Explanation: SYN scans send SYN and wait for response but don't complete handshake.

23.

Why are stealth scans preferred over full connect scans?

- A) They are faster
- B) They avoid detection by security devices
- C) They discover fewer open ports
- D) They crash the target

Answer: B

Explanation: Stealth scans reduce the chance of detection.

24.

Which of these is NOT typically part of a security evaluation plan?

- A) Scope
- B) Rules of engagement
- C) Exploit development
- D) Reporting methods

Answer: C

Explanation: Exploit development is a hacking activity, not part of planning.

25.

Traceroute uses which protocol primarily?

- A) TCP
- B) ICMP or UDP
- C) HTTP
- D) FTP

Answer: B

Explanation: Traceroute commonly uses ICMP or UDP packets.

26.

In vulnerability scanning, what is a false positive?

- A) A correctly identified vulnerability
- B) A vulnerability that is not actually present
- C) A successful exploit
- D) A denied port

Answer: B

Explanation: False positives indicate vulnerabilities that don't really exist.

27.

What is the main drawback of active scanning?

- A) It is slow
- B) It can be detected by the target
- C) It does not find vulnerabilities
- D) It is illegal

Answer: B

Explanation: Active scanning interacts with the target and can trigger alarms.

28.

Which scan type sends FIN packets to a target?

- A) SYN scan
- B) FIN scan
- C) NULL scan
- D) XMAS scan

Answer: B

Explanation: FIN scan sends FIN flags to probe ports.

29.

What is the first step in ethical hacking?

- A) Exploitation
- B) Footprinting
- C) Reporting
- D) Scanning

Answer: B

Explanation: Footprinting/recon is the initial information gathering step.

30.

Which of the following is a limitation of footprinting?

- A) It always provides complete information
- B) It requires physical access
- C) It may not reveal internal network details
- D) It crashes the network

Answer: C

Explanation: Footprinting mainly collects public info and may miss internal data.

☒ Hard (10 Questions)

31.

What technique does an IDLE scan rely on to perform a stealthy port scan?

- A) Direct scanning from attacker IP

- B) Spoofing IP addresses and using a “zombie” host’s IP ID sequence
- C) Using social engineering
- D) Brute forcing passwords

Answer: B

Explanation: IDLE scans exploit predictable IP ID values of a zombie host.

32.

Why might a NULL scan bypass some firewalls?

- A) It completes the TCP handshake
- B) It sends no TCP flags, making packets appear anomalous and sometimes ignored
- C) It floods the network
- D) It uses encryption

Answer: B

Explanation: Some firewalls do not expect packets with no flags and ignore them.

33.

Which scanning method can cause denial of service if overused?

- A) Passive scanning
- B) Aggressive active scanning
- C) Stealth scanning
- D) NULL scanning

Answer: B

Explanation: Aggressive scanning floods the target with packets, possibly causing DoS.

34.

How does a SYN scan identify open ports?

- A) Open ports respond with SYN/ACK packets
- B) Open ports send RST packets
- C) Open ports drop the packets
- D) Open ports never respond

Answer: A

Explanation: Open ports respond with SYN/ACK to SYN requests.

35.

Which is NOT a common reason ethical hackers perform port scanning?

- A) Identify open services
- B) Find potential entry points
- C) Exploit buffer overflows
- D) Map the attack surface

Answer: C

Explanation: Exploiting overflows comes after scanning, not part of scanning itself.

36.

Which of these is a legal concern when performing footprinting?

- A) Using publicly available info
- B) Unauthorized probing or scanning
- C) Using search engines
- D) Gathering domain info

Answer: B

Explanation: Unauthorized scans can be illegal and unethical.

37.

Which scan tries to avoid detection by not completing the TCP handshake?

- A) Full connect scan
- B) SYN scan
- C) Ping scan
- D) UDP scan

Answer: B

Explanation: SYN scans are half-open and less detectable.

38.

What is the purpose of a vulnerability scanner's database?

- A) Store known vulnerabilities and signatures
- B) Store passwords
- C) Store network maps
- D) Store user data

Answer: A

Explanation: Scanner databases contain info on known vulnerabilities.

39.